Exercise 1: Eddington-Finkelstein Coordinates

A general static, spherically symmetric metric can be written as

$$ds^2 = -A(r)\,dt^2 + \frac{dr^2}{B(r)} + r^2 d\Omega^2, \tag{1}$$

where $d\Omega^2$ is the metric on a unit 2-sphere. Assume that $A(r)$ and $B(r)$ are smooth functions of r such that both have a simple zero at $r = r_+ > 0$ and are positive for $r > r_+$.

a)* Show that radial null geodesics are given by $t \pm r_* =$ constant, where

$$r_* = \int_{r_0}^{r} \frac{dx}{\sqrt{A(x)B(x)}}, \tag{2}$$

with $r_0 > r_+$ an arbitrary constant. Show that $r_* \to -\infty$ as $r \to r_+$.

$$ds^2 = -A(r)dt^2 + \frac{dr^2}{B(r)} + \overset{\text{radial}}{0} = \overset{\text{null}}{0}$$

$$A(r)dt^2 = \frac{dr^2}{B(r)}$$

$$dt = \pm \frac{dr}{\sqrt{A(r)B(r)}}$$

$$t = \pm \int_{r_0}^{r} \frac{dx}{A(x)B(x)} + C$$

As desired.

b) We substitute $t = v - r_*$, $dt = dv - \frac{dr_*}{dr}dr$, $dt^2 = dv^2 + \left(\frac{dr_*}{dr}\right)^2 dr^2 - \frac{dr_*}{dr}dv\,dr$

$$ds^2 = -A(r)\,dt^2 + \frac{dr^2}{B(r)} + r^2 d\Omega^2$$

$$= -A(r)\left[dv^2 + \left(\frac{dr_*}{dr}\right)^2 dr^2 - 2\frac{dr_*}{dr}dv\,dr\right] + \frac{dr^2}{B(r)} + r^2 d\Omega^2$$

$$= -A(r)dv^2 + 2\sqrt{\frac{A(r)}{B(r)}}\,dv\,dr - \frac{1}{B(r)}dr^2 + \frac{dr^2}{B(r)} + r^2 d\Omega^2$$

$$= -A(r)dv^2 + 2\sqrt{\frac{A(r)}{B(r)}}\,dv\,dr + r^2 d\Omega^2$$

Now $g \xrightarrow{r \to r_+}$ some finite value, the only term where this was really in question was the square root but it's fine since both zeroes are simple, thus we can analytically continue it.

Exercise 2: Acceleration of Killing Orbits

In a spacetime that possesses a timelike Killing vector field k, consider trajectories of massive particles that move along orbits of k, that is, their four-velocity is a unit-normalized timelike vector u that is proportional to k.

a)* Show that the acceleration of these particles, $a^\mu = u^\nu \nabla_\nu u^\mu$, can also be written as

$$a_\mu = \nabla_\mu \log \sqrt{-k_\nu k^\nu}\,. \tag{3}$$

Can you now identify a function $\phi(x)$ that can be interpreted as the gravitational potential in which the particle moves?

We want $u_\nu u^\nu = -1$

Hence $u^\mu = -|k_\nu k^\nu|^{-1/2} k^\mu = -\|k\|^{-1} k^\mu$

$$a^\mu = u^\nu \nabla_\nu u^\mu$$

$$= -\frac{1}{\|k\|} k^\nu \nabla_\nu \left(-\frac{1}{\|k\|} k^\mu\right)$$

$$= \frac{1}{\|k\|} k^\nu \left[\frac{1}{\|k\|}\nabla_\nu k^\mu + k^\mu \nabla_\nu\left(\frac{1}{\|k\|}\right)\right]$$

$$= \frac{1}{\|k\|} k^\nu \left[\frac{1}{\|k\|}\nabla_\nu k^\mu + k^\mu\left(-\frac{1}{\|k\|^2}\right)\nabla_\nu(\|k\|)\right]$$

$$a_\mu = \frac{1}{\|k\|} k^\nu \left[\frac{1}{\|k\|}\nabla_\nu k_\mu + k_\mu\left(-\frac{1}{\|k\|^2}\right)\nabla_\nu(\|k\|)\right]$$

Killing equation

$$\nabla_\mu k_\nu + \nabla_\nu k_\mu = 0$$

$$a_\mu = \frac{1}{\|k\|} k^\nu \left[-\frac{1}{\|k\|}\nabla_\mu k_\nu + k_\mu\left(-\frac{1}{\|k\|^2}\right)\nabla_\nu(\|k\|)\right]$$

$$= -\frac{1}{\|k\|^2}\left[\frac{1}{2}\nabla_\mu(k_\nu k^\nu) + \frac{1}{\|k\|} k_\mu k^\nu \nabla_\nu\left(\sqrt{-k_\sigma k^\sigma}\right)\right]$$

Now we show the second term vanishes. To do so, we can show

$$k^\nu \nabla_\nu (k^\mu k_\mu) \stackrel{?}{=} 0$$

$$k^\nu \nabla_\nu (k^\mu k_\mu) = 2 k^\nu k^\mu \nabla_\nu k_\mu$$

$$= -2 k^\nu k^\mu \nabla_\mu k_\nu$$

$$= -k^\rho \nabla_\nu (k^\gamma k_\gamma)$$

Hence $k^\gamma \nabla_\gamma (k^\nu k_\nu) = 0$

Thus $a_\nu = -\frac{1}{||k||^2} \frac{1}{2} \nabla_\nu (k_\gamma k^\gamma)$

$$= \frac{1}{2} \nabla_\nu \ln(-k_\gamma k^\gamma)$$

$$= \nabla_\nu \ln\sqrt{-k_\gamma k^\gamma}$$

The gradient of potential is force

$$a_\nu = -\nabla_\nu \phi$$

Thus $\phi = -\ln\sqrt{-k_\gamma k^\gamma}$

b)* Compute a^μ for particles moving along orbits of $k = \partial_t$ in the Schwarzschild metric

$$ds^2 = -\left(1 - \frac{2M}{r}\right) dt^2 + \frac{dr^2}{1 - \frac{2M}{r}} + r^2 d\Omega^2 . \tag{4}$$

What kind of situation do these trajectories represent? Obtain the gravitational potential ϕ. Can you interpret these results?

$$-k_\gamma k^\gamma = 1 - \frac{2M}{r}$$

$$a_\nu = \nabla_\nu \ln\sqrt{1 - \frac{2M}{r}}$$

Clearly the only nonzero component is a_r

$$a_r = \frac{1}{2} \partial_r \ln\left(1 - \frac{2M}{r}\right)$$

$$= \frac{1}{2} \frac{1}{1 - \frac{2M}{r}} \left(\frac{2M}{r^2}\right)$$

Since g is diagonal, only a^r is nonzero

$$a^r = \left(1 - \frac{2M}{r}\right) a_r$$

$$= \frac{M}{r^2}$$

The gravitational potential is $\phi = -\frac{1}{2}\ln\left(1 - \frac{2M}{r}\right)$

$$\simeq -\frac{M}{r}, \quad \frac{M}{r} \ll 1$$

This agrees with the newtonian result. The acceleration agrees too, and is that of a free falling observer.

c)* Evaluate a^μ as $r \to 2M$. Can you identify this result? Read Harvey's 9.2.

Harvey evaluates $||a||$ which makes sense since a^ν is [illegible]

$$a^2 = g_{rr} a^r a^r = \frac{1}{1 - \frac{2M}{r}} \left(\frac{M}{r^2}\right)^2$$

$$a \xrightarrow{r \to 2M} \infty$$

Exercise 3: Surface Gravity Identities

Let $\mathcal{H}$ be a Killing horizon with KVF ξ^μ. We will assume that ξ^μ is timelike on one side of $\mathcal{H}$ (the "exterior" of the black hole, connected to asymptotic infinity). Let us also define $|\xi| := \sqrt{-\xi_\mu \xi^\mu}$ and the surface gravity of the Killing horizon $\mathcal{H}$ as the non-affinity of ξ^μ,

$$\xi^\mu \nabla_\mu \xi^\nu =: \kappa \xi^\nu . \tag{5}$$

a) Use Frobenius' Theorem to prove

$$\kappa^2 = -\frac{1}{2} (\nabla^\mu \xi^\nu)(\nabla_\mu \xi_\nu)|_{\mathcal{H}} . \tag{6}$$

Frobenius's theorem $\xi_{[\nu} \nabla_\gamma \xi_{\rho]} = 0$

$$0 = \xi_\nu \nabla_\gamma \xi_\rho + \xi_\rho \nabla_\nu \xi_\gamma + \xi_\gamma \nabla_\rho \xi_\nu$$

$$= \xi_\nu \nabla_\gamma \xi_\rho + \xi_\rho \nabla_\nu \xi_\gamma - \xi_\gamma \nabla_\nu \xi_\rho$$

$$\xi_\rho \nabla_\nu \xi_\gamma = \xi_\gamma \nabla_\nu \xi_\rho - \xi_\nu \nabla_\gamma \xi_\rho$$

Choose null vector N^μ with $N^\mu \xi_\mu = -1$

$$N^\rho \xi_\rho \nabla_\mu \xi_\nu = \xi_\nu N^\rho \nabla_\mu \xi_\rho - \xi_\mu N^\rho \nabla_\nu \xi_\rho$$

$$-\nabla_\mu \xi_\nu = \xi_\nu N^\rho \nabla_\mu \xi_\rho - \xi_\mu N^\rho \nabla_\nu \xi_\rho$$

Defining $\omega_\mu = N^\rho \nabla_\mu \xi_\rho$,

$$\nabla_\mu \xi_\nu = \xi_\mu \omega_\nu - \xi_\nu \omega_\mu$$

Now, $\nabla_\mu(\xi^\nu \xi_\nu) = 2\xi^\nu \nabla_\mu \xi_\nu$

$$= -2\xi^\nu \nabla_\nu \xi_\mu$$
$$= -2\kappa \xi_\mu$$

$$(\nabla_\mu \xi^2)(\nabla^\mu \xi^2) = -2\kappa \xi^2$$

$$\xi^\nu \xi^\rho (\nabla_\mu \xi_\nu)(\nabla^\mu \xi^\rho) = -2\kappa \xi^2$$

Taking this limit gives (c) but idk about (a) ⍨

b) Prove that $\xi^\mu \nabla_\mu |\xi| = 0$.

$$|\xi| = \sqrt{-\xi^2}$$

$$\nabla_\mu \xi^2 = -2\kappa \xi_\mu$$

$$\nabla_\mu |\xi| = -\frac{\kappa}{|\xi|} \xi_\mu$$

$$\xi^\mu \nabla_\mu |\xi| = -\frac{\kappa}{|\xi|} \xi^2$$
$$= 0 \text{ on } \mathcal{H}.$$

Exercise 4: Traversable Wormholes

The geometry

$$ds^2 = -dt^2 + d\rho^2 + (\rho^2 + a^2)\left(d\theta^2 + \sin^2\theta\, d\phi^2\right), \qquad (8)$$

where a is a real constant $-\infty < \rho < +\infty$, is a simple model of a traversable wormhole. You should convince yourself of this interpretation.

Consider a congruence of radial null geodesics in this geometry, with $\xi = \partial_t - \partial_\rho$ as its generator.

a)* Argue that this vector describes null rays that are ingoing in the region $\rho > 0$. Are they also ingoing in the region $\rho < 0$?

They are obviously null. Since $\frac{d\rho}{dt} = -1$, they are only going in for $\rho > 0$.

b)* Argue that the vorticity of this congruence is zero. Conclude that the shear vanishes as well. The more efficiently you obtain these results, the better.

The vorticity is 0 because ξ is hypersurface orthogonal (to the surface $t - r = \text{const.}$)

c)* Compute the expansion θ of the congruence (again, be efficient). Interpret the result: is the congruence always converging? Then compute the rate of change of the expansion $\xi^\mu \partial_\mu \theta$.

$$\theta = B_{\mu\nu} g^{\mu\nu}$$

$$B_{\mu\nu} = -\Gamma^\lambda_{\mu\nu} k_\lambda$$

$$= \Gamma^t_{\mu\nu} + \Gamma^r_{\mu\nu}$$

However, the metric is independent of t

$$B_{\mu\nu} = \Gamma^r_{\mu\nu}$$

Thus, $\Gamma^r_{\theta\theta} = -r$, $\Gamma^r_{\phi\phi} = -r\sin^2\theta$

$$\theta = -\frac{2r}{r^2 + a^2}$$

$$\xi^\mu \nabla_\mu \theta = \xi^r \partial_r \theta$$

$$= \frac{2(a^2 - r^2)}{(r^2 + a^2)^2}$$

d)* Insert the previous results into Raychaudhuri's equation to obtain the null-null component $\xi^\mu \xi^\nu T_{\mu\nu}$ of the stress-energy tensor that would be required in order that this geometry solves the Einstein Equations. What can you say about energy conditions in this spacetime?

That guy's equation (null congruences)

$$\frac{d\theta}{d\lambda} = -\frac{1}{D-2}\theta^2 - \cancelto{0}{\sigma_{\mu\nu}\sigma^{\mu\nu}} + \cancelto{0}{w_{\mu\nu}w^{\mu\nu}} - R_{\mu\nu}\xi^\mu\xi^\nu$$

EFE: $G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}$

$$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu}$$

Since ξ is null, by definition $g_{\mu\nu}\xi^\mu\xi^\nu = 0$

$$R_{\mu\nu}\xi^\mu\xi^\nu = \kappa T_{\mu\nu}\xi^\mu\xi^\nu$$

$$R_{\mu\nu}\xi^\mu\xi^\nu = -\frac{1}{D-2}\theta^2 - \frac{d\theta}{d\lambda}$$

$$= -\frac{2(a^2 - r^2)}{(r^2 + a^2)^2} - \frac{1}{2}\left[\frac{2r}{r^2 + a^2}\right]^2$$

$$= -\frac{1}{(r^2 + a^2)^2}\left[2a^2 - 2r^2 + 2r^2\right]$$

$$= -\frac{2a^2}{(r^2 + a^2)^2}$$

$T_{\mu\nu}\xi^\mu\xi^\nu = -\frac{1}{\kappa}\frac{2a^2}{(r^2+a^2)^2} < 0$, violating NEC (hence WEC & SEC)